

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and rCommerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

Roll No. :53	Name:Sakshi Vipin Yadav
Class : SYIT	Batch : 1
Date of Assignment :15/7/26	Date/Time of Submission : 15/7/26

a. Write a program to: a. Implement a queue using an array.

Code:

```

from array import*

que = array('i', [])

def enqueue(data):
    que.append(data)
    print("Data inserted:", data)

def dequeue():
    if len(que) == 0:
        print("Queue is empty")
    else:
        print("Data removed:", que.pop(0))

def peek():
    if len(que) == 0:
        print("Queue is empty")
    else:
        print("First element is:", que[0])

def show():
    print("Queue:", que)

enqueue(10)
enqueue(20)
enqueue(30)

show()
peek()

dequeue()
show()
print("Yadav Sakshi Vipin S053")

```

Output:

```

Data inserted: 10
Data inserted: 20
Data inserted: 30
Queue: array('i', [10, 20, 30])
First element is: 10
Data removed: 10
Queue: array('i', [20, 30])
Yadav Sakshi Vipin S053

```

- a. Simulate a simple queuing system (e.g., customer service queue).

```

import time

que = [("P1", 2), ("P2", 5), ("P3", 2), ("P4", 0)]

print("Running order")

print(que)

while que:
    job, timer = que.pop(0)
    print(f"Running: {job} {timer} sec")
    time.sleep(timer)

print(que)

|
print("Yadav Sakshi Vipin S053")

```

Output;

```

REDIRIT: C:/Users/ANVILG/App
Running order
[('P1', 2), ('P2', 5), ('P3', 2),
Running: P1 2 sec
Running: P2 5 sec
Running: P3 2 sec
Running: P4 0 sec
[]
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```